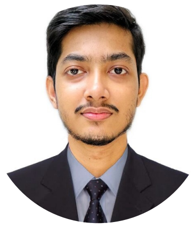
MD MAHIR UDDIN

Data Scientist | Machine Learning & Gen AI

mahiruddin.vercel.app | mahir.uddin.0@gmail.com | +880 185 999 0434 | Dhaka, Bangladesh

LinkedIn | GitHub | Codeforces | LeetCode | Kaggle | Coursera



ABOUT ME

I am a Data Scientist with strong foundations in Mathematics, Statistical Modeling, Deep Learning, and Computer Vision. Driven by a long-standing love for Mathematics and problem-solving, I transitioned into the field of Data Science by pursuing an MSc in CSE (Data Science). Currently working on an ADB-funded project that uses AI and IoT to increase right first time shade in bulk dyeing and reduce resource usage in textile dyeing factories. Worked on image detection generation, LLM fine-tuning, and RAG, with ongoing research in developing a density-based clustering algorithm and modification of ADAM optimizer for reduced oscillation and faster convergence.

SKILLS

- **Programming Languages:** Python, SQL, C, C++, JavaScript
- **Machine Learning & AI:** Machine Learning, Deep Learning, Computer Vision, Generative AI
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, LangChain
- **MLOps/Deployment:** FastAPI, Rest APIs, Docker, AWS EC2
- **Others:** Git, Linux, HTML, CSS MS Excel, LaTeX
- **Additional Knowledge:** Big Data, Cloud Computing, Data Engineering

PROJECTS

Skin Diseases Prediction with LLM Recommendations

Vision Transformer, LLM, FastAPI, Docker, PyTorch

- Built a Data efficient image Transformer (DeiT)-based skin disease classifier (97%+ accuracy) with real-time inference via FastAPI.
- Integrated LLM(Gemini 2.5 Flash) driven recommendations and deployed using Docker on the cloud with an interactive streamlit interface.

Portrait Image to Sketch Translation with Pix2Pix GAN

TensorFlow, Pix2Pix GAN, U-Net, PatchGAN

- Developed a Pix2Pix conditional GAN with U-Net generator and PatchGAN discriminator for photo-to-sketch translation on FS2K dataset.
- Generated high-fidelity, structurally accurate sketches using adversarial and L1 loss for paired image-to-image translation.

LLM Fine-Tuning with RAG for Investment Guidance

LLM, LangChain, Vector DB, RAG

- Fine-tuned a domain-specific LLM and integrated Retrieval-Augmented Generation to deliver context-aware investment insights.
- Combined model fine-tuning with dynamic retrieval from books, annual reports, and financial articles from the web.

Stock Market Trend Prediction using ANFIS

ANFIS, PyTorch, Scikit-learn

- Implemented an Adaptive Neuro-Fuzzy Inference System (ANFIS) to automatically learn indicator patterns that signal stock price uptrends.
- Achieved 76.8% accuracy in predicting price trend of the next day.

More projects: mahiruddin.vercel.app/projects

ACHIEVEMENTS

- Codeforces max rating **1302**; ranked **773rd** among **20,772** Bangladeshi coders (**Top 2.98%**).
- International Youth Math Challenge (IYMC) 2025 Gold Medalist.
- Ranked **14th nationwide** in BUTEX A Unit 2018 admission exam among **13,000+** candidates.
- Ranked **9th nationwide** in Jahangirnagar University H Unit 2018 admission exam among **34,000+** candidates.

RESEARCH

Research works completed as part of coursework and independent study (unpublished; publishable work not included).

Automated Density-Based Splitting of Merged Clusters

Clustering, Unsupervised Learning — Paper

- Proposed a clustering refinement framework to detect and split incorrectly merged clusters using density analysis
- Designed a recursive splitting mechanism to dynamically adjust cluster structure without predefined *k*.

Fabric Defect Detection Using Histogram Equalization and CNN

Computer Vision, CNN — Paper

- Developed a vision-based textile defect detection system combining contrast enhancement with CNNs.
- Improved defect visibility and classification robustness under varying fabric textures and lighting conditions.

Intelligent Irrigation Decision Support System Using IoT and Weather Data

IoT, Data Analytics — Paper

- Built a data-driven decision support system integrating IoT sensor data with weather information.
- Optimized irrigation planning to reduce water consumption and enhance resource efficiency, with motor operation time and duration controlled via IoT microcontrollers.

EDUCATION

M.Sc. in Computer Science & Engineering
United International University
Nov 2024–June 2026 (Expected) Dhaka, Bangladesh

- Major: Data Science
- CGPA: 3.82

B.Sc. in Textile Engineering
Bangladesh University of Textiles (BUTEX)
Jan 2018–Oct 2023 Dhaka, Bangladesh

- Specialization: Wet Processing Engineering
- CGPA: 3.34

REFERENCES

Dr. Ohidujjaman
Associate Professor, Department of CSE
United International University

- Email: ohidujjaman@cse.uiu.ac.bd
- Phone: +880 179 808 3444

Dr. Mohammad Nurul Huda
Professor, Dept. of CSE
United International University

- Email: mnh@cse.uiu.ac.bd
- Phone: +880 1673-476433

EXPERIENCE

Research Engineer (AI/ML) *(Part-time)*
ADB-Funded AI & IoT Project
Feb 2026 – Present

Dhaka, Bangladesh

Mathematics & Physics Lecturer *(Part-time)*
Sunrise Coaching Centre
Aug 2018 – Sep 2022 (4 yrs 2 months)

Dhaka, Bangladesh

Mathematics Lecturer *(Seasonal)*
UCC (University Coaching Center)
Apr 2019 – Aug 2022 (3 yrs 5 months)

Dhaka, Bangladesh

CERTIFICATIONS

- **Machine Learning Specialization**
  Stanford · Coursera Credentials
- **Deep Learning Specialization**
  DeepLearning.AI · Coursera Credentials
- **IBM Generative AI Engineering Professional Certificate**
  IBM · Coursera Credentials
- **Mathematics for Machine Learning and Data Science**
  DeepLearning.AI · Coursera Credentials